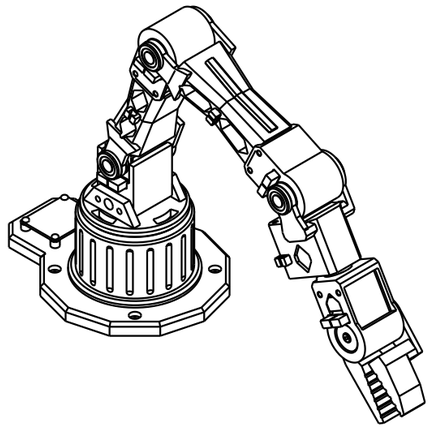
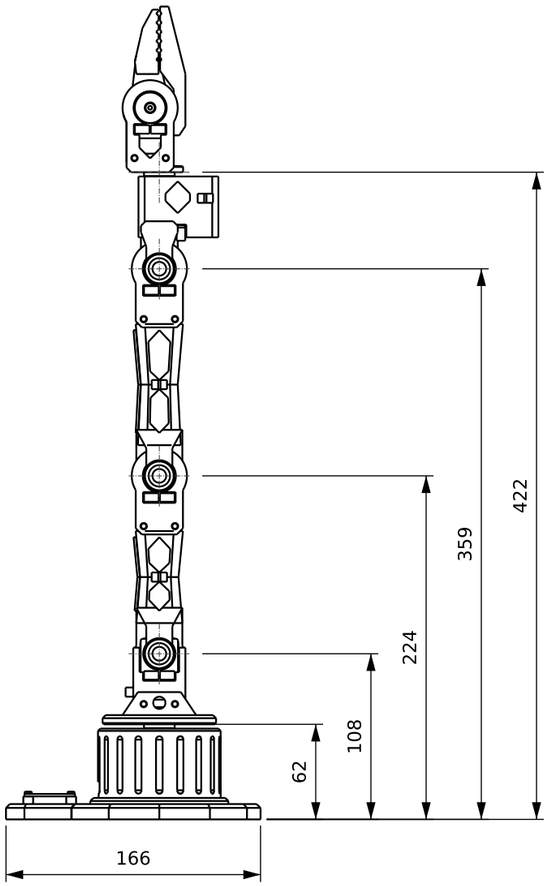


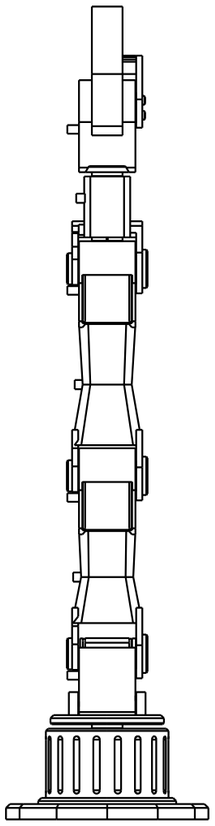
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

PARTS LIST (set = work set, A or B)

1.	TAL-101 Base	1	PLA	set A
2.	TAL-102 Shoulder (turntable+cradle)	1	PLA	set A
3.	TAL-103 Upper arm	1	PLA	set A
4.	TAL-104 Forearm	1	PLA	set B
5.	TAL-105 Wrist	1	PLA	set B
6.	TAL-106 Gripper body	1	PLA	set B
7.	TAL-107 Moving jaw	1	PLA	set B
8.	TAL-108 Hub cap, large	3	PLA	set A
9.	TAL-109 Hub cap, small	3	PLA	set A
10.	Feetech STS3215 servo (12 V)	6	bought	
11.	Waveshare Bus Servo Adapter (A)	1	bought	
12.	M3 x 6 button head (servo horns)	24	bought	
13.	M3 x 8 button head (passive horns)	3	bought	
14.	M2 x 5 self-tapping (servo cases)	12	bought	
15.	M2 x 6 self-tapping (servo cases)	8	bought	
16.	M2.5 x 6 self-tapping (board)	4	bought	
17.	M4 bolts to the table (user's)	4	bought	
18.	3-pin bus servo cables	6	with servos	

JOINT LAYOUT

1. Pan axis: vertical through the base centre. Pan horn face 62 above the table.
2. Shoulder axis (along Y): 46 above the pan horn face, on the pan axis.
3. Upper arm 116, forearm 135 (axis to axis). Wrist flex axis to roll horn face 63. Jaw pivot 42 ahead of the roll horn face, 6 above the roll axis.
4. Clevis plates on every pitch joint: output plate +Y (y 18.9..22.9), idler plate -Y (y -22.9..-18.9).
5. Front view: zero pose (all joints 0, arm straight up). Isometric: 'ready' pose.

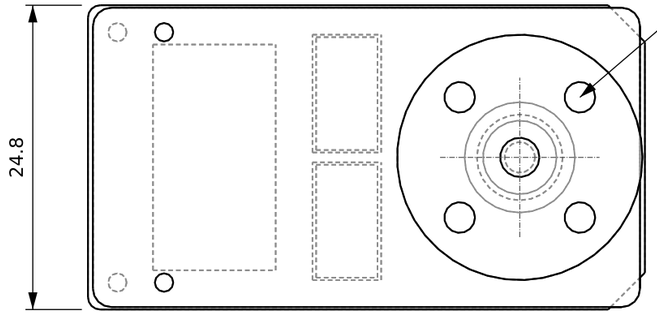
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part sheets give each part in its own frame; this sheet shows where those frames sit.

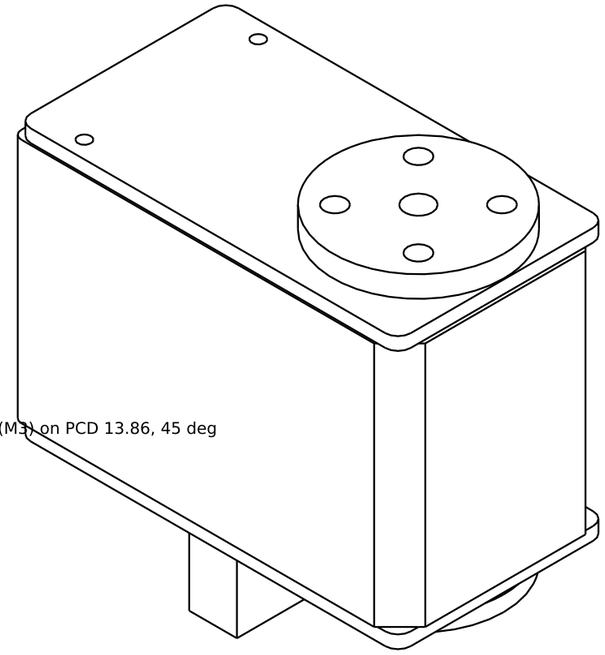
REFERENCE: in work sets A and B

THIRD ANGLE

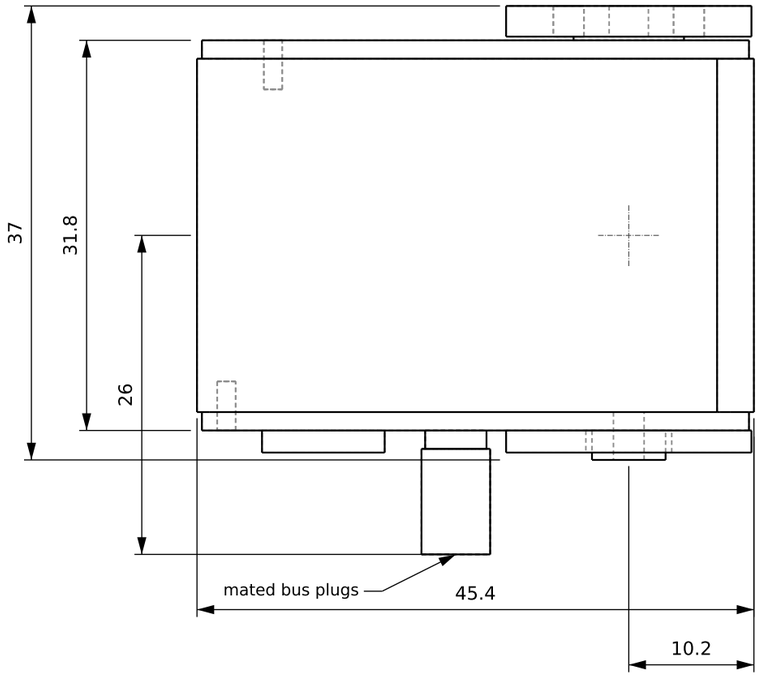
PROJECT TALOS 5-axis arm	EST. MASS -
PART **General assembly and parts list**	SCALE 1:4
PART NO. **TAL-000**	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-03
MATERIAL -	SHEET 1 of 12



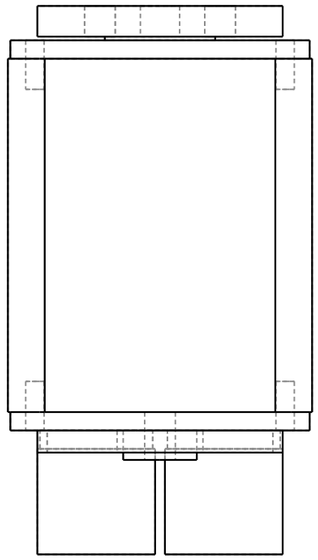
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

FEATURES (servo frame)

1. Body: box x -35.2..10.2, y -12.4..12.4, z -14.4..14.4 (45.4 x 24.8 x 28.8). +X end vertical corners chamfered 3, -X end corners R1.
2. Caps: 44.6 x 24.4 x 1.5, corners R1.5, centred on x = -12.5, at z 14.4..15.9 and -15.9..-14.4.
3. Back pad (connector face): x -29.9..-19.9, y -9.2..9.2, z -17.7..-15.9.
4. Bus connector headers: x -16.6..-11.6, y 0.6..9.8 and -9.8..-0.6, z -17.4..-15.9. Mated plugs: x -16.9..-11.3, y 0.4..10.0 and -10.0..-0.4, z -26.0..-17.4.
5. Spline boss dia 9, z 14.7..16.2. Idler boss dia 6, z -18.3..-14.4, tapped dia 2.5 (M3).
6. Output horn dia 20, z 16.2..18.7: 4x dia 2.5 (M3) on PCD 13.86 at 45/135/225/315 deg, centre dia 3.2.
7. Passive horn dia 20, z -17.7..-15.9, centre dia 7.
8. Case holes dia 1.5 x 4 deep (M2 self-tapping): shaft face at (x, y) = (-29.0, +/-10.2); connector face at (-32.8, +/-10.2).

   SERVO POCKET P (cut this into a part, in the servo frame; 0.3 clearance included):

   a) box x -35.5..10.5, y -12.7..12.7, z -16.2..16.2

   b) back pad box x -30.2..-19.6, y -9.5..9.5, z -18.0..-15.5

   c) plug bay box x -17.6..-10.6, y -10.6..10.6, z -28.5..-15.5

   d) idler clearance cylinder dia 21.6, z -18.5..-15.0, on the shaft axis

   e) horn clearance cylinder dia 21.6, z 15.0..19.5, on the shaft axis

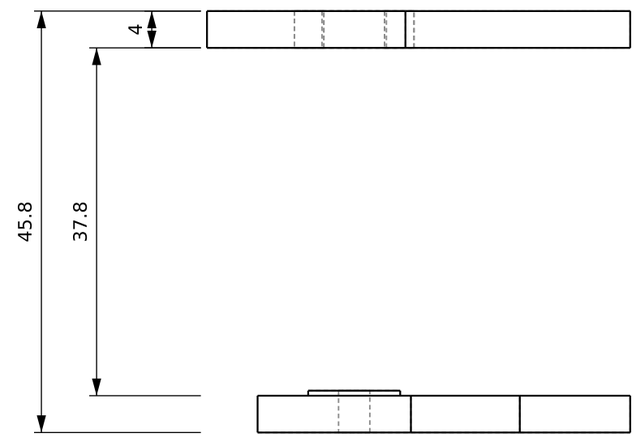
NOTES

- Measured from the SO-ARM100 STS3215 reference model and Feetech's drawing. Check the connector position and case-hole depth on a real servo.
- Each part sheet says where its pocket's servo frame sits and how it is rotated.

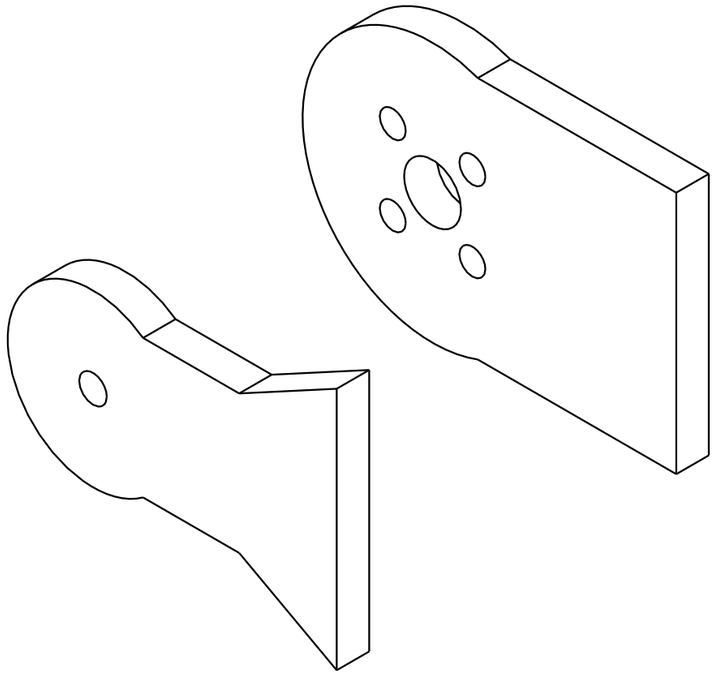
REFERENCE: in work sets A and B

THIRD ANGLE

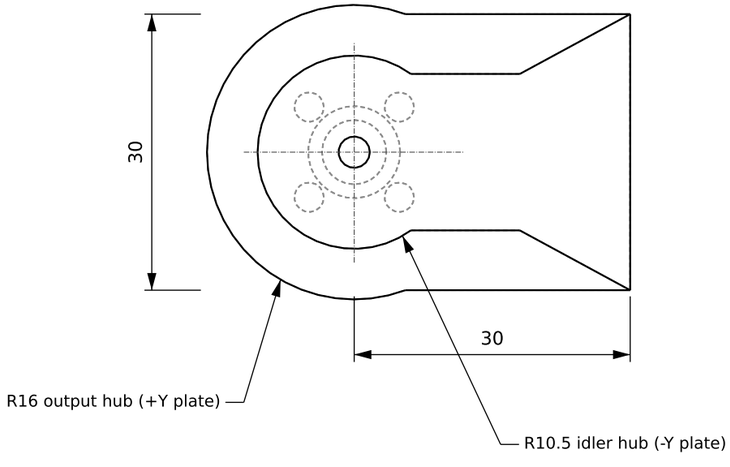
PROJECT TALOS 5-axis arm	EST. MASS -
PART **STS3215 servo + pocket envelope (reference)**	SCALE 2:1
PART NO. **TAL-001**	UNITS / REV mm / A
QTY / ARM 6	DATE 2026-11-03
MATERIAL bought part	SHEET 2 of 12



TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

COMMON FEATURES

CLEVIS (every pitch joint; drawn for the arm links: reach R = 30, half-height H = 15; wrist uses R = 24, H = 12). Plates are 4 thick, normal to Y.

1. Output plate, y 18.9..22.9: circle R16 at the origin + rectangle x 0..R, z -H..H.
2. Output plate holes, through: centre dia 7 (horn spline screw access) + 4x dia 3.2 at (x, z) = (+/-4.9, +/-4.9) (PCD 13.86, 45 deg).
3. Idler plate, y -22.9..-18.9: circle R10.5 + neck polygon (0,10.5) (6,8.5) (0.6R,8.5) (R,H) (R,-H) (0.6R,-8.5) (6,-8.5) (0,-10.5). The neck clears the parent servo's bus plugs.
4. Idler plate: pad dia 10 on the inner face, y -18.9..-18.35; centre hole dia 3.4 through.
5. Clevis gap in the link body: remove a cylinder R21 about Y between y -18.9..18.9, and everything at x < 0 between those planes.

CASE SCREW HOLE (servo case, M2 self-tapping): dia 2.4 from the servo face to the seat, dia 4.2 counterbore from the seat outward. Seat = distance from the servo mid-plane: arm shells and gripper 17.3, shoulder 18.5, base 18.4, wrist 18.0.

HORN SCREW PATTERN: 4x on PCD 13.86 at 45/135/225/315 deg + centre. Clearance dia 3.2 (M3); counterbore dia 6.5 where the head sits inside a part.

CABLE CLIP (one bus cable; local u = along cable, v = across, w = out of the face): block u -3..3, v -5..5, w -1..5 (1 mm sunk into the face); tunnel u through, v -2.5..2.5, w 0.8..4.3; slit v -0.75..0.75, w 4.2..6; chamfer 1 on the two outer long edges.

HEX WINDOW H(xa, xb, zc, k): hexagon (xa,zc) (xa+k,zc-k) (xb-k,zc-k) (xb,zc) (xb-k,zc+k) (xa+k,zc+k), cut through the part along Y.

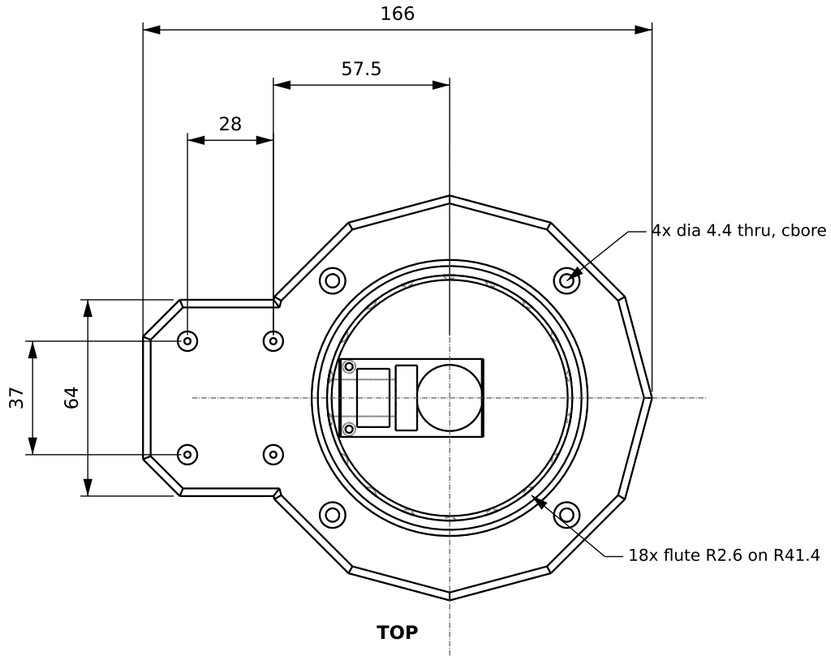
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.

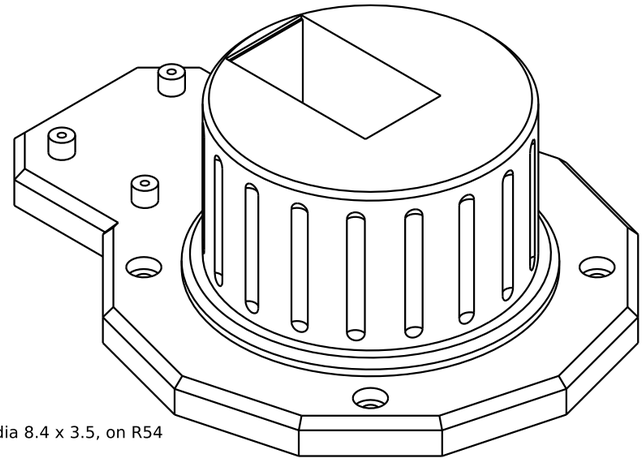
REFERENCE: in work sets A and B

THIRD ANGLE

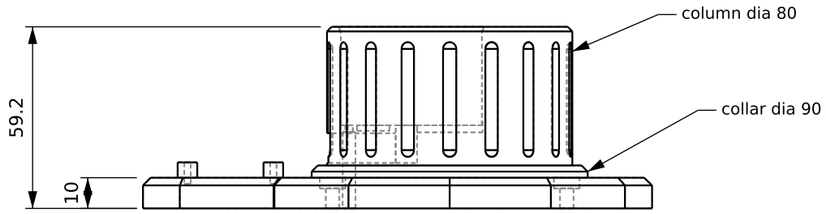
PROJECT TALOS 5-axis arm	EST. MASS -
PART **Common features: clevis, clip, screw holes**	SCALE 3:2
PART NO. **TAL-002**	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 3 of 12



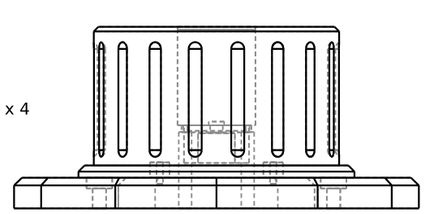
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

FEATURES (in order)

1. Plinth: regular 12-gon, circumscribed dia 132 (a corner on +X), extrude z 0..10.
2. Board deck: polygon (-40,-32) (-88,-32) (-100,-20) (-100,20) (-88,32) (-40,32), extrude z 0..10; union with the plinth.
3. Chamfer 2.5 on all top outer edges (z = 10).
4. Mounting holes: 4x dia 4.4 through on R54 at 45/135/225/315 deg; counterbore dia 8.4 from z 6.5 up.
5. Collar: cylinder dia 90, z 10..14; chamfer 2 on its top edge.
6. Column: cylinder dia 80, z 10..59.2; chamfer 1.5 on its top edge.
7. Board standoffs: 4x cylinder dia 6.4, z 9..15, at (-57.5,+/-18.5) and (-85.5,+/-18.5); pilot dia 2.1 from z 8 to 16 (M2.5 self-tapping).
8. Flutes: 18x vertical cylinder dia 5.2 on R41.4 at 10 deg + 20 deg steps, z 19..52, with dia 5.2 spheres at both ends; cut.
9. Servo pocket P (sheet TAL-001) with the servo frame origin at (0, 0, 43.3); servo +Z (shaft) -> +Z, servo +X -> +X, servo +Y -> +Y.
10. Top opening: cut box x -36..11, y -12.7..12.7, z 58.2..64.2.
11. Cable tunnel (plug bay -> board deck): cut box x -60..-13, y -6..6, z 14.8..24.5.
12. Case screw holes (sheet TAL-002): connector face only, screws from below: (x, y) = (-32.8, +/-10.2); hole z 24.9..27.5, counterbore z -1.7..24.9 (opens on the bottom). Each is a dia 2.4 hole from the servo face to the seat, then a dia 4.2 counterbore from the seat outward through the part.

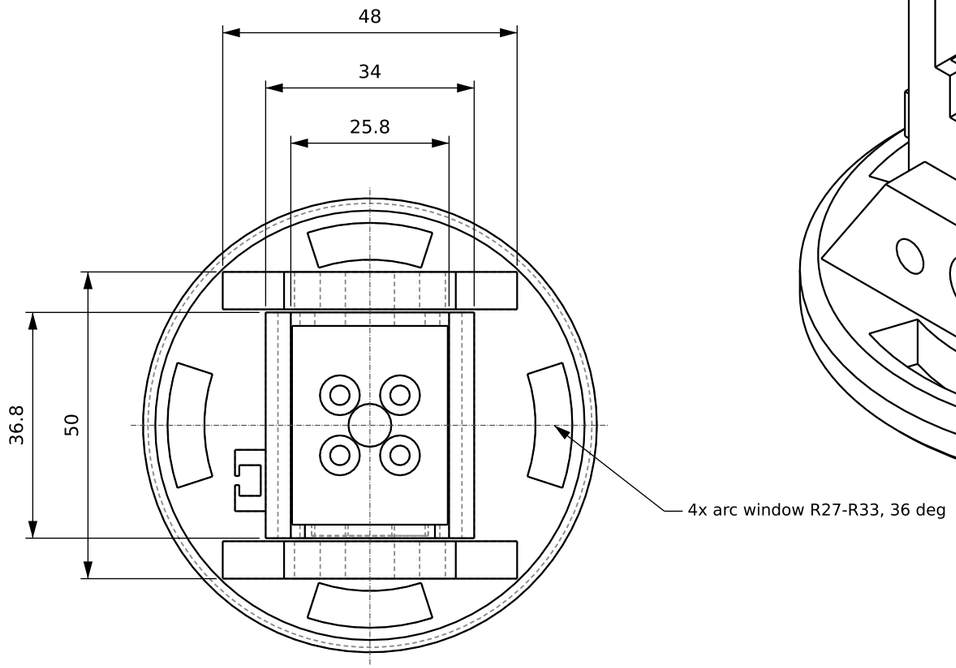
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The pan servo drops in from the top; its plugs hang into the bay under it.

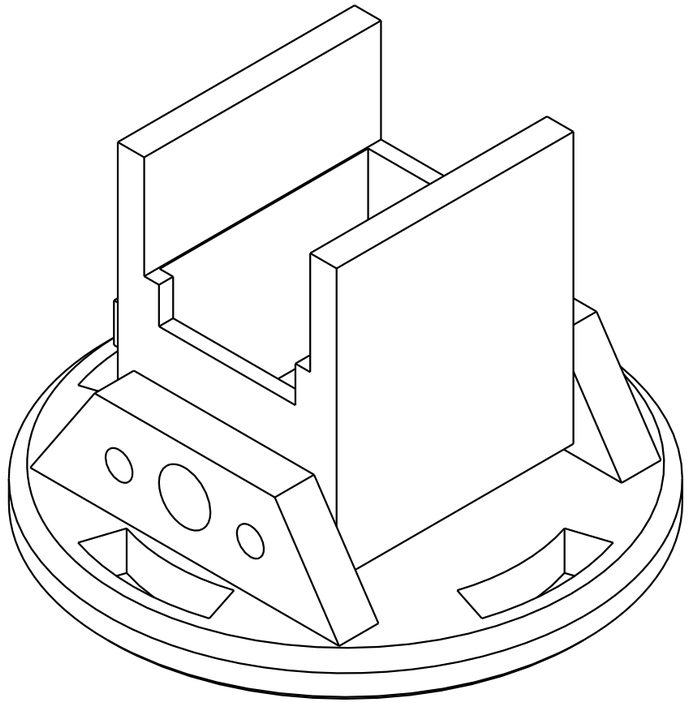
WORK SET A: base, shoulder, upper arm, hub caps

THIRD ANGLE

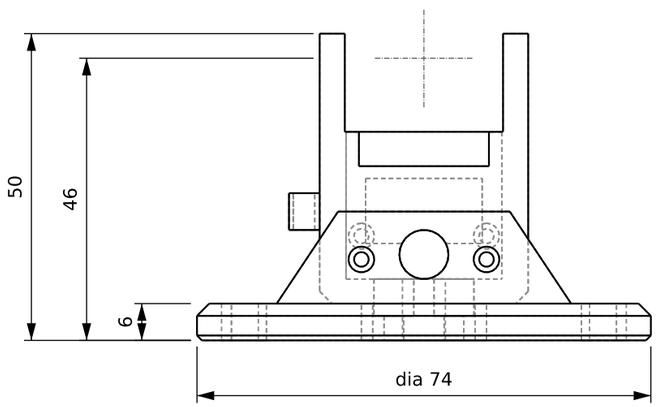
PROJECT TALOS 5-axis arm	EST. MASS 155 g
PART **Base**	SCALE 1:2
PART NO. **TAL-101**	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 4 of 12



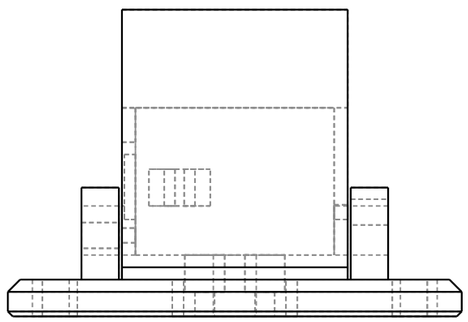
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

FEATURES (in order)

1. Turntable: cylinder dia 74, z 0..6; chamfer 2 top edge, 0.8 bottom edge.
2. Cradle: box x -17..17, y -18.4..18.4, z 5..34; chamfer 3 on the 4 edges parallel to Y.
3. Posts: boxes x -17..-12.9 and 12.9..17, y -18.4..18.4, z 30..50.
4. Cheeks: trapezoid (-24,6) (24,6) (14,21) (-14,21) in XZ, extruded y 18.9..25 and y -25..-18.9.
5. Cable clip (TAL-002) on the back face: centre (-17, -9, 21), cable along Z, face normal -X.
6. Servo pocket P (sheet TAL-001) with the servo frame origin at (0, 0, 46); servo +Z (shaft) -> +Y, servo +X -> +Z (body hangs down), servo +Y -> +X.
7. Servo opening: cut box x -12.7..12.7, y -16.2..16.2, z 10..80.
8. Cable hole: dia 8 along Y, through all, at x = 0, z = 14.
9. Pan horn: centre dia 7, z -1..11. 4x dia 3.2 at (+/-4.9, +/-4.9), z -1..4, counterbore dia 6.5 z 4..11 (M3 x 6 from inside the cradle).
10. Turntable windows: 4x arc slot between R27 and R33, -18..+18 deg about 0/90/180/270 deg, through.
11. Case screw holes (sheet TAL-002): seat 18.5, through the cheeks: shaft face at (x, z) = (+/-10.2, 17), y +15.9..+18.5; connector face at (+/-10.2, 13.2), y -18.5..-15.9. Each is a dia 2.4 hole from the servo face to the seat, then a dia 4.2 counterbore from the seat outward through the part.

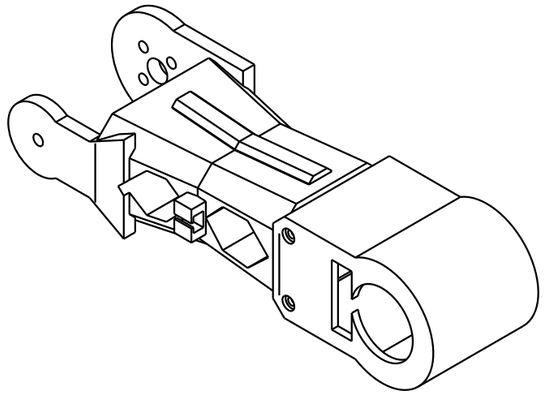
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.

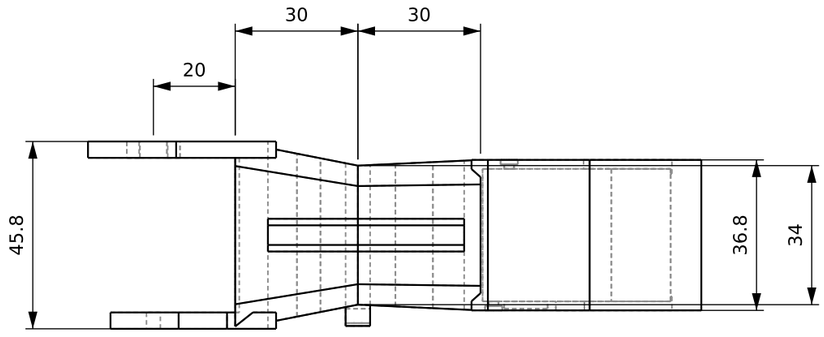
WORK SET A: base, shoulder, upper arm, hub caps

THIRD ANGLE

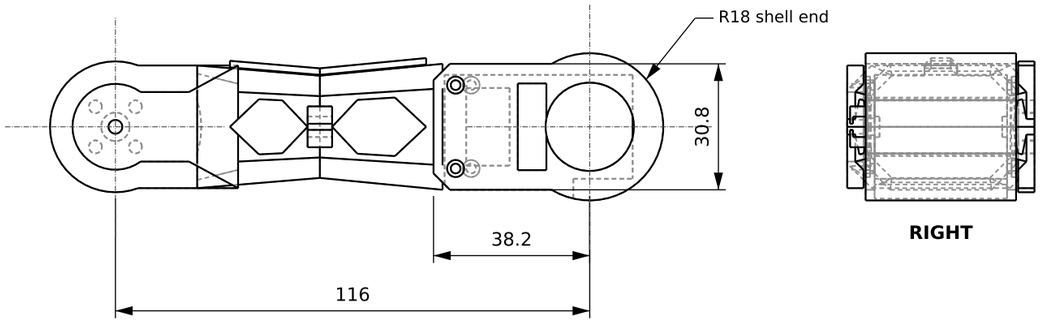
PROJECT TALOS 5-axis arm	EST. MASS 39 g
PART **Shoulder (turntable + cradle)**	SCALE 1:1
PART NO. **TAL-102**	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 5 of 12



ISOMETRIC (NOT TO SCALE)



TOP



FRONT

RIGHT

FEATURES (in order)

1. Clevis (TAL-002), reach 30, half-height 15.
2. Beam: ruled loft of YZ sections, x 20: 45.8 wide x 30 tall, corners chamfered 6; x 50: 34 wide x 25 tall, corners chamfered 5; x 80: 36.8 wide x 30.8 tall, corners chamfered 6.
3. Servo shell: XZ polygon (81.8,-15.4) (116,-15.4) (116,15.4) (81.8,15.4) (77.8,11.4) (77.8,-11.4), extruded y -18.4..18.4; plus a cylinder R18 about Y at x = 116, y -18.4..18.4.
4. Spine: ruled loft of the ridge profile through the same three stations, then trimmed to x 28..76. Ridge profile at beam height h: (-4,h/2-3) (4,h/2-3) (4,h/2+0.2) (2.4,h/2+1.8) (-2.4,h/2+1.8) (-4,h/2+0.2).
5. Window 1, cut through Y: (28,0) (35.1,6.75) (40.85,6.46) (47,0) (40.85,-6.46) (35.1,-6.75).
6. Window 2, cut through Y: (53,0) (59.19,6.58) (68.35,7.16) (76,0) (68.35,-7.16) (59.19,-6.58).
7. Cable clip (TAL-002) on the -Y flank at the waist: centre (50, -17.2, 0), cable along X.
8. Clevis gap cut (TAL-002).
9. Servo pocket P (sheet TAL-001) with the servo frame origin at (116, 0, 0); servo +Z (shaft) -> +Y, servo +X -> +X, servo +Y -> -Z.
10. Servo opening (drops in from below): cut box x 80.5..112, y -16.2..16.2, z -40..-12.
11. Case screw holes (sheet TAL-002): seat 17.3; shaft face (+Y) at (x, z) = (87, +/-10.2); connector face (-Y) at (83.2, +/-10.2). Each is a dia 2.4 hole from the servo face to the seat, then a dia 4.2 counterbore from the seat outward through the part.

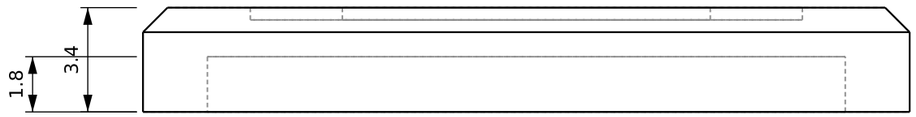
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The elbow servo's bus plugs come out through the -Y shell wall.

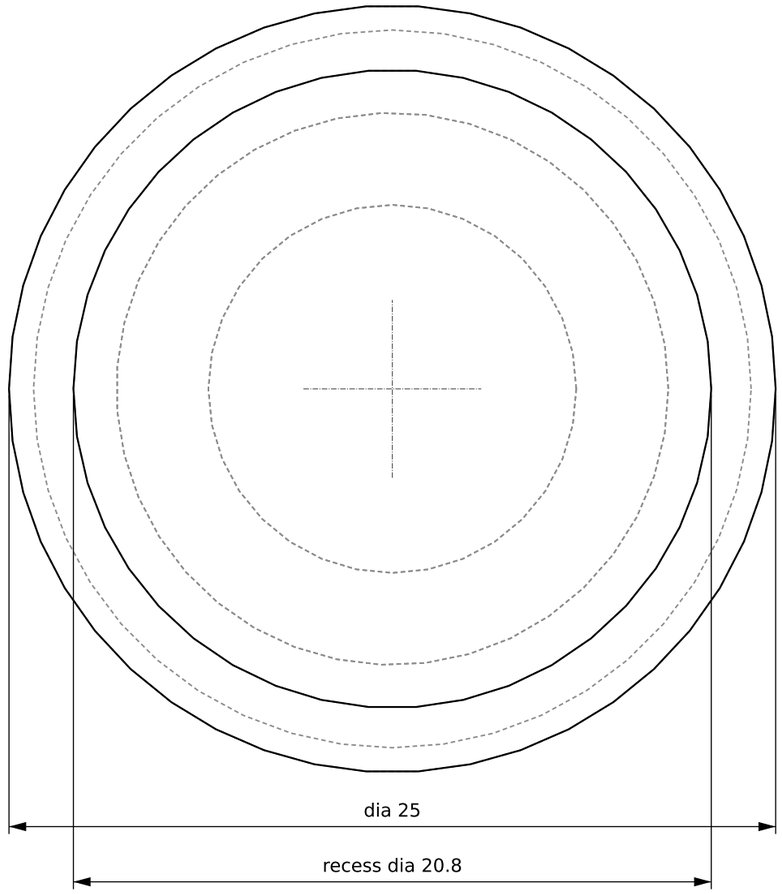
WORK SET A: base, shoulder, upper arm, hub caps

THIRD ANGLE

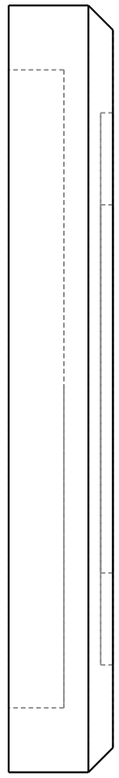
PROJECT TALOS 5-axis arm	EST. MASS 51 g
PART **Upper arm**	SCALE 2:3
PART NO. **TAL-103**	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 6 of 12



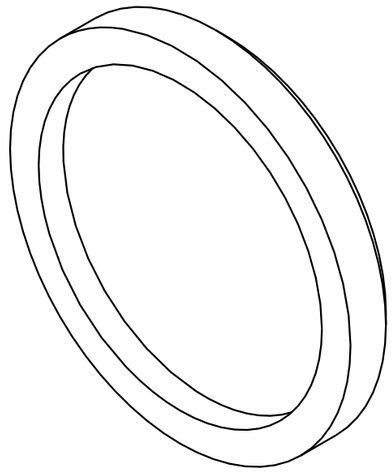
TOP



FRONT



RIGHT



ISOMETRIC (NOT TO SCALE)

FEATURES

HUB CAP, LARGE (output side, covers 4 horn screws):

1. Disc dia 25, y 0..3.4; chamfer 0.8 on the outer edge.
2. Recess dia 20.8, y 0..1.8 (clears M3 button heads, 1.65 high).
3. Face groove: annulus between dia 12 and dia 18, 0.4 deep (y 3.0..3.4).

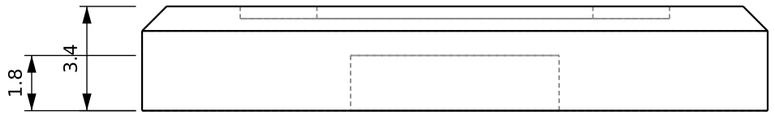
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Press-fit or glue onto the clevis plate over the screw heads. Print face down.

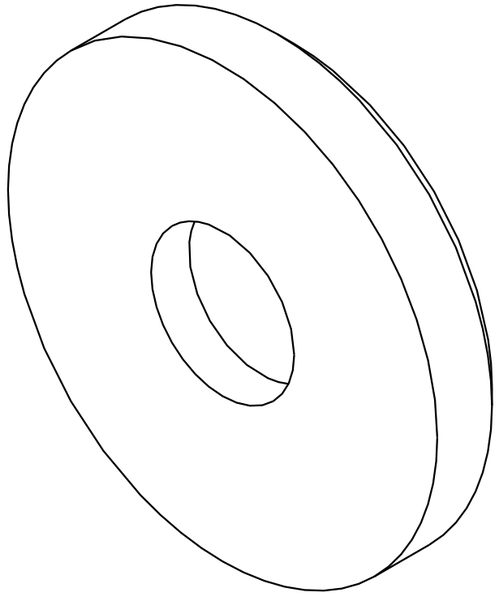
WORK SET A: base, shoulder, upper arm, hub caps

THIRD ANGLE

PROJECT TALOS 5-axis arm	EST. MASS 1 g
PART **Hub cap, large**	SCALE 5:1
PART NO. **TAL-108**	UNITS / REV mm / A
QTY / ARM 3	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 11 of 12



TOP



ISOMETRIC (NOT TO SCALE)

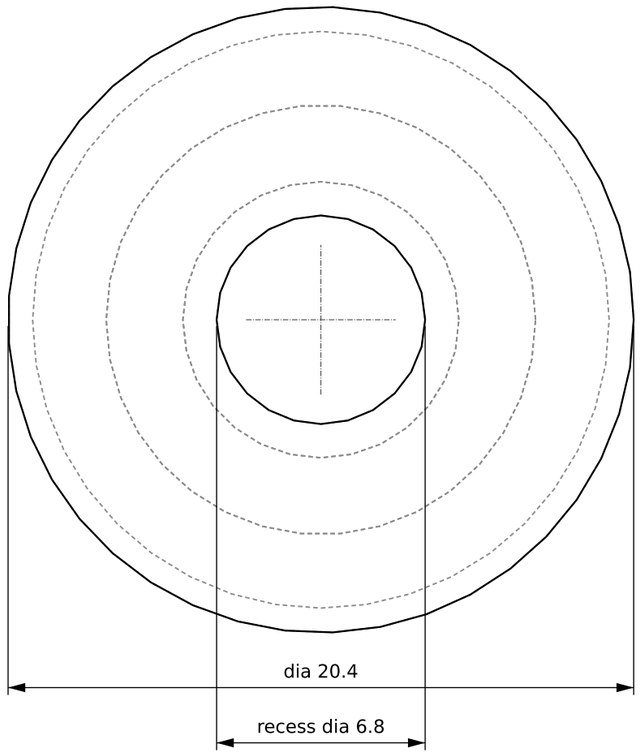
FEATURES

HUB CAP, SMALL (idler side, covers 1 centre screw):

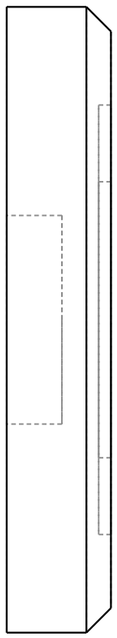
1. Disc dia 20.4, y 0..3.4; chamfer 0.8 on the outer edge.
2. Recess dia 6.8, y 0..1.8 (clears M3 button heads, 1.65 high).
3. Face groove: annulus between dia 9 and dia 14, 0.4 deep (y 3.0..3.4).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Press-fit or glue onto the clevis plate over the screw heads. Print face down.



FRONT



RIGHT

WORK SET A: base, shoulder, upper arm, hub caps

THIRD ANGLE

PROJECT TALOS 5-axis arm	EST. MASS 1 g
PART **Hub cap, small**	SCALE 5:1
PART NO. **TAL-109**	UNITS / REV mm / A
QTY / ARM 3	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 12 of 12